

Automatic Fan Controller For Air-conditioners



RAJ K. GORKHALI

Many central air-conditioning systems (ACs) have the option for cooling and heating both. These have a fan for blowing hot or cold air drawn from a central unit, often with no automatic speed control for the fan. The speed of the AC fan has to be manually controlled to maintain a comfortable temperature throughout the room.

Manually adjusting the AC temperature is not very convenient. So we present here an automatic control system for the AC fan which could also help control the room temperature to some extent. It allows the air-conditioning system to automatically blow warm air in winters and cool air in summers without requiring manual intervention.

Circuit description

Fig. 1 shows the circuit of the automatic fan controller for the AC. It comprises regulator IC 7805 (IC1), bar-graph driver IC LM3914 (IC2), comparator

IC LM339 (IC3), temperature sensor IC LM335 (IC4) and some discrete components. Pin configurations of 7805, BC337 and LM335 are shown in Fig. 2.

Power for the circuit is derived from mains supply. The 230V, 50Hz AC mains is stepped down by transformer X1 to deliver a secondary output of 12V, 500 mA. The transformer output is rectified by a full-wave rectifier comprising diodes D1 through D4, filtered by capacitor C6 and regulated by IC 7805. Capacitor C7 bypasses ripples in the regulated supply.

IC LM3914 is configured for bargraph mode by connecting its pin 9 to 5V supply. It functions both as the temperature and set-point indicator, depending on whether switch S1 is in RUN or SET position, respectively. A highly stable internal reference voltage is generated at pin 7 using resistor combination R1-R2 and presets VR1 and VR2. This voltage is directly fed to the upper end of the inter-

nal voltage divider chain at pin 6. Pin 4, the lower end of the internal voltage divider chain, is connected to the wiper of preset VR1. The voltage difference between pins 4 and 6 determines the range of temperature control.

Potentiometer VR3 connected between pins 4 and 6 of IC2 provides temperature setting depending on the position of switch S1. The comparator inside IC2 compares the voltage at pin 5 with the voltage difference across pins 4 and 6, and incrementally turns on LED1 through LED10 at every tenth of the temperature range. Current driven through the LEDs is regulated and programmable, thus eliminating the need for resistors.

The temperature control function is performed by comparator IC LM339.

This IC uses only three of the four independent precision comparators operating off a single power supply. Comparator A1 is wired as a non-inverting comparator with hysteresis. R6 is used

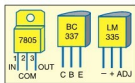


Fig. 2: Pin configurations of 7805, BC337 and LM335

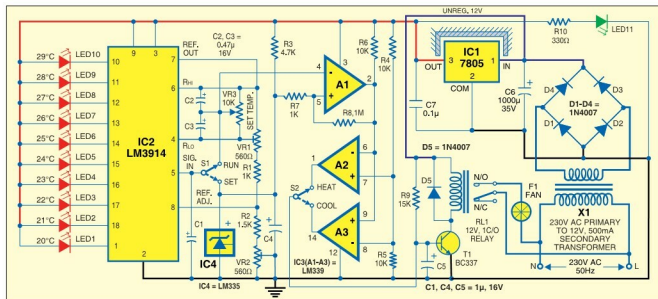


Fig. 1: Circuit of the automatic fan controller for ACs